

# Predictive Modeling Using Machine Learning

## Salary Prediction

**Tools:** Python, Pandas, Scikit-learn, Matplotlib, Seaborn

**Data:** Cleaned salary dataset from Task 1 (1,783 rows; Age, Gender, Education Level, Job Title, Years of Experience, Salary)

### 1. Objective

Build supervised learning models that predict outcomes from data. **Part A (regression)** predicts the exact Salary. **Part B (classification)** predicts whether a salary is High or Low (High means at or above the median of \$110,000), which allows confusion matrix and ROC curve evaluation.

### 2. Method

1. Used the cleaned data from Task 1. Duplicates were already removed, so identical rows cannot appear in both the training and test sets (which would give falsely high scores).
2. Categorical columns (Gender, Education Level, Job Title) were one-hot encoded inside a scikit-learn Pipeline.
3. Split the data 80% train (1,426 rows) and 20% test (357 rows), random\_state = 42.
4. Trained Linear Regression, Decision Tree (max depth 8) and Random Forest (300 trees). Also ran 5-fold cross-validation to check that the scores are stable.
5. Part B: trained Logistic Regression and Random Forest classifiers on the High / Low salary label.

### 3. Part A: Regression Results

| Model                | R2 (test) | MAE (\$) | RMSE (\$) | CV R2 (5-fold) |
|----------------------|-----------|----------|-----------|----------------|
| Linear Regression    | 0.824     | 15,738   | 21,883    | 0.427          |
| Decision Tree        | 0.820     | 15,735   | 22,092    | 0.627          |
| Random Forest (best) | 0.893     | 11,264   | 17,070    | 0.697          |

Random Forest is the best model: on unseen test data its predictions are off by about \$11,000 on average, and it explains about 89% of salary variation. The cross-validated R2 (0.70) is lower than the single test score, so the realistic expectation for new data is around 0.70 to 0.89.

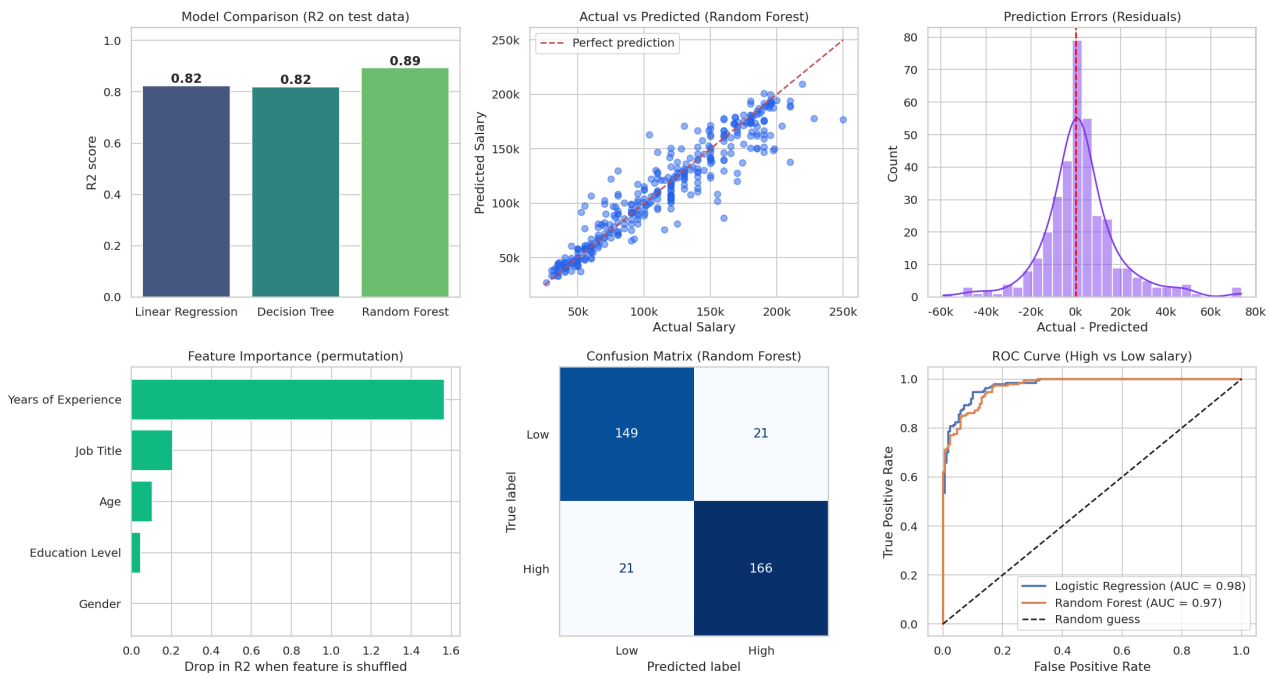
### 4. Part B: Classification Results (High vs Low salary)

| Model               | Accuracy | Precision | Recall | F1    | AUC   |
|---------------------|----------|-----------|--------|-------|-------|
| Logistic Regression | 0.905    | 0.918     | 0.898  | 0.908 | 0.976 |
| Random Forest       | 0.882    | 0.888     | 0.888  | 0.888 | 0.971 |

Both classifiers correctly label about 9 out of 10 employees. The Random Forest confusion matrix shows 149 Low and 166 High predicted correctly, with 21 mistakes in each direction. AUC above 0.97 means the models separate high and low earners very well.

## 5. Visual Results

### Predictive Modeling Results - Salary Dataset



## 6. Key Insights

- Years of Experience is by far the most important feature**, followed by Job Title, Age and Education Level.
- Gender has almost no effect on the prediction** once experience and role are known. The raw gender gap seen in Task 1 is therefore mostly explained by differences in experience and job roles.
- Random Forest beats Linear Regression and Decision Tree because it captures non-linear patterns (salary growth slows after about 20 years of experience).
- About 83% of test predictions are within \$20,000 of the real salary. Rare job titles are slightly harder to predict (average error about \$12.7K versus \$11.1K for common titles).

## 7. Conclusion

Supervised learning can predict salary reliably from experience, role and education. Random Forest gave the best regression accuracy ( $R^2 = 0.89$ , MAE about \$11K), and the classifiers reached about 90% accuracy with AUC near 0.98. More data on rare job titles would improve the predictions further.